Affect as Precision Augmentation for Social Inference: Per-Partner Metacognitive States in Multi-Agent Active Inference

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Abstract

We present a computational model of affect as per-partner metacognitive precision tracking in multi-agent active inference. Each social partner is assigned a slow-timescale affective state β_k that summarizes the running reliability of the agent's predictive model for that partner. This state augments policy evaluation by precision-weighting expected free energy estimates with partner-specific confidence information. In a volatile multi-partner trust game, we show that: (1) affect provides *orthogonal* augmentation beyond explicit planning depth—under sophisticated inference, non-affective planners from horizon $\tau = 2$ to $\tau = 8$ are statistically indistinguishable, yet adding the affective signal yields $d = 0.64$ improvement at every depth; (2) ablating affect reproduces the somatic marker deficit—intact partner knowledge with impaired decision deployment; (3) precision tracking and reward averaging occupy distinct informational niches, with precision tracking excelling under betrayal stress ($d = 0.59$) and reward averaging prevailing in continuous-investment settings ($d = 0.43$–0.89 over precision tracking); (4) Bayesian model comparison decisively favors the affective model under volatility ($\log_{10}$ BF $= 3.0$); (5) augmentation generalizes across Prisoner's Dilemma, Stag Hunt, and Chicken games ($d > 1.0$ under betrayal in all three); and (6) clinical parameter perturbations (alexithymia, borderline, depression) produce qualitatively distinct behavioral phenotypes across environments—in the Stag Hunt, borderline is the first phenotype with decisive Bayesian evidence of impairment ($d = -0.72$, $\log_{10}$ BF $= -2.89$), while alexithymia is paradoxically protective. The mechanism occupies a distinct computational niche from both outside-in empathy and cognitive theory of mind: inside-out metacognitive monitoring of one's own social inference channel. The architecture transfers without modification to a spatial multi-agent benchmark (Cogs vs. Clips), where the same beta update rule tracks teammate positional predictability. Affect encodes information orthogonal to both planning depth and reward history: partner-specific model fitness that improves social decisions through a channel inaccessible to explicit deliberation alone.

Keywords: active inference, affect, social cognition, precision weighting, somatic markers, multi-agent systems, computational psychiatry

1 Introduction

Social cognition presents a fundamental computational bottleneck. When an agent interacts with multiple social partners whose strategies may change unpredictably, optimal decision-making requires maintaining probabilistic models of each partner's hidden states, simulating forward through multiple rounds of interaction, and evaluating expected outcomes under candidate policies. The computational cost grows exponentially with planning horizon, yet humans navigate these environments fluently—suggesting they employ computational shortcuts that preserve decision quality without exhaustive forward planning.

Damasio's somatic marker hypothesis [Damasio, 1994] proposes that emotional signals provide exactly this shortcut. Through the ventromedial prefrontal cortex (vmPFC) and its connections to the amygdala and insula, the brain generates rapid affective evaluations that bias decision-making before and below deliberate reasoning. The Iowa Gambling Task [Bechara et al., 1994, 1997] provides the key evidence: vmPFC-lesioned patients can verbally identify risky options but fail to develop the anticipatory somatic markers that healthy participants use to guide advantageous choices. The dissociation—intact knowledge but impaired deployment—is the behavioral signature our model aims to reproduce computationally in a social context.

Active inference [Friston, 2010, Parr et al., 2022] provides the formal framework. Agents maintain generative models of observation causes and act to minimize variational free energy. Policy selection is governed by expected free energy (EFE), decomposing into pragmatic value (preference alignment) and epistemic value (uncertainty reduction). Hesp et al. [2021] formalized affect within this framework: valence encodes expected model precision (subjective fitness), with affective charge driven by signed prediction errors. Critically, their model operates in a single-agent context. What is absent is a per-partner metacognitive affective state that compresses interaction history into a precision signal for multi-agent social inference.

Existing multi-agent active inference work covers theory of mind as inference over others' generative models [Friston & Frith, 2015, Yoshida et al., 2008], empathy as model coupling [Pitliya et al., 2025, Matsumura et al., 2024], and factorized multi-agent models [Ruiz-Serra et al., 2025]. These approaches define three distinct computational strategies for social cognition. *Outside-in empathy* [Pitliya et al., 2025] couples agents' generative models so that one agent's affective or perceptual states directly inform another's inference—the computational direction flows from partner to self. *Cognitive theory of mind* [Yoshida et al., 2008] maintains an explicit model of the partner's inference process, recursively simulating their beliefs and policies. The present work introduces a third strategy: *inside-out metacognitive monitoring*, where the agent tracks the reliability of its own predictive model for each partner, without modeling the partner's internal states or coupling to their generative process. This triple dissociation—outside-in empathy, cognitive ToM, and inside-out precision monitoring—carves the space of social inference mechanisms at its computational joints. None of the existing approaches provides a per-partner metacognitive affective state that: (a) compresses interaction history into a precision signal, (b) provides orthogonal augmentation beyond what explicit planning depth recovers, and (c) whose ablation reproduces the somatic marker deficit pattern.

Contributions. We introduce three novel components:

1. **Per-partner metacognitive affective state in multi-agent active inference.** Not a global mood or shared precision parameter, but a learned state for each social partner compressing that specific interaction history into a precision signal.
2. **Affect as orthogonal augmentation beyond planning depth.** Under sophisticated inference, explicit planning depth ($\tau = 2$ through $\tau = 8$) produces identical non-affective performance; the affective signal adds evaluative information that no amount of additional lookahead recovers.
3. **Computational vmPFC lesion in a social context.** The ablation reproduces intact social knowledge with impaired deployment, extending the IGT literature to multi-agent settings.

2 Model

2.1 Three-Level Architecture

The model has three hierarchical levels operating on different timescales:

Level 1—Observations (per-trial). Trial-by-trial interaction outcomes: partner's action, own payoff. Updated every round.

Level 2—Cognitive social model (parameter learning). Latent states encoding each partner's type/strategy (cooperator, reciprocator, exploiter, random). Dirichlet parameters accumulate evidence over many rounds.

Level 3—Per-partner affective state (metacognitive integration). A slow state β_k for each partner k summarizing the running quality of the agent's social model. Updated by an exponentially-weighted moving average of prediction error magnitudes, resistant to single-trial fluctuations.

2.2 Generative Model

The agent maintains a POMDP generative model for each social partner k:

$$P(o_{1:T}, s_{1:T}, \pi) = P(\pi) \prod_{t=1}^{T} P(o_t \mid s_t)\, P(s_t \mid s_{t-1}, \pi) \tag{1}$$

with hidden states factoring hierarchically: $s_t = (s_t^{(1)}, s_t^{(2)}, s_t^{(3)})$ for trial state, partner type, and affective state respectively.

State inference exploits the analytical solution to variational free energy minimization: for a categorical $q(s)$, the VFE-minimizing posterior equals the Bayesian posterior $q^*(s) \propto P(o \mid s)P(s)$. The implementation uses one-step Bayes with the likelihood matrix $\mathbf{A}$ and transition matrix $\mathbf{B}$.

2.3 Expected Free Energy and Policy Selection

For policy π at future timestep t:

$$G_t(\pi) = \underbrace{-\mathbb{E}_{Q(o_t|\pi)}[\ln P(o_t)]}_{\text{pragmatic value}} - \underbrace{\mathbb{E}_{Q(s_t|\pi)}[D_{\mathrm{KL}}[Q(s_t|o_t,\pi)\|Q(s_t|\pi)]]}_{\text{epistemic value}} \tag{2}$$

Policy selection follows a softmax: $P(\pi) = \sigma(-\gamma \cdot G(\pi))$, where γ is the policy precision. The planner uses observation-branching sophisticated inference across all conditions, enumerating future binary observation sequences, updating beliefs pathwise via Bayes rule, and averaging pathwise EFE under path probabilities.

2.4 Affective State Dynamics

The affective state β_k for partner k is updated from a signed affective charge derived from prediction error:

$$\beta_k^{(t+1)} = \lambda\, \beta_k^{(t)} + (1 - \lambda)\, \sigma\big(\phi(\epsilon_k^{(t)})\big) \tag{3}$$

where $\lambda \in (0.5, 0.9)$ is a smoothing parameter, $\epsilon_k^{(t)} = 1 - P(o_t^{\mathrm{obs}} \mid s_k^{(2)})$ is the unsigned surprise from the level-2 model, and $\phi(\epsilon) = \alpha(\sigma_0^2 - \epsilon^2)$ converts surprise into affective charge. When actual squared surprise is below baseline σ_0^2, affect increases (model is accurate); when above, it decreases (model is failing).

This update extends Hesp et al.'s variational treatment of precision dynamics to the per-partner social setting, building on the free-energy formalization of emotional valence by Joffily

& Coricelli [2013] and the interoceptive inference framework of Seth [2013]. The surprise term is computed from the agent's own level-2 predictive beliefs, making the affective update belief-coupled rather than externally injected.

2.5 Affect as Precision Weighting

The affective agent augments standard EFE with a partner-specific precision weight:

$$G(\pi) \approx \left(\sum_{t=1}^{\tau} G_t(\pi)\right) \cdot (1 + \mu\,\beta_k) \tag{4}$$

where μ scales the influence of affect on policy selection. The weight is keyed to the first partner implicated by the policy so the signal survives first-action marginalization.

2.6 The vmPFC Lesion

The computational lesion sets $\mu = 0$: the agent "feels" things about partners (β_k updates normally) but feelings have no effect on action selection. This models a disconnection between metacognitive affective representation and decision circuitry, preserving intact level-2 posteriors while impairing behavioral deployment.

2.7 Reward-Average Control (Condition 5)

To distinguish precision tracking from cached value, a reward-average agent replaces β_k with a simple exponential moving average of reward:

$$\bar{R}_k^{(t+1)} = \lambda\,\bar{R}_k^{(t)} + (1-\lambda)\,r_t, \qquad \tilde{R}_k = \tfrac{1}{2}(1 + \tanh(\bar{R}_k/\max|r|)) \tag{5}$$

The terminal signal $\tilde{R}_k$ enters the same precision-weighting mechanism as β_k, enabling controlled comparison of precision-based versus value-based augmentation.

2.8 Discrete Variational Beta

The discrete variational formulation casts β_k as a discrete hidden state with $L = 5$ precision levels, explicit likelihood $P(\epsilon \mid \beta) \propto \exp(\alpha(\sigma_0^2 - \epsilon^2) \cdot \beta)$, and tridiagonal persistent transition dynamics. The posterior update uses the same categorical Bayesian machinery as partner-type inference:

$$q(\beta_k^{(t)}) \propto P(\epsilon_k^{(t)} \mid \beta_k) \sum_{\beta_k^{(t-1)}} P(\beta_k^{(t)} \mid \beta_k^{(t-1)})\, q(\beta_k^{(t-1)}) \tag{6}$$

In stable conditions, continuous EMA and discrete formulations are indistinguishable ($d = 0.001$). Under betrayal, the discrete formulation underperforms moderately ($d = 0.41$) due to the transition matrix constraining single-step posterior shifts—a feature reflecting a stronger prior on precision stability.

3 Experimental Design

3.1 Multi-Partner Trust Game

The agent interacts with $K = 4$ partners over $N = 200$ rounds. Each round, a partner is selected (randomly or by agent choice), both choose cooperate or defect simultaneously, and payoffs follow a Prisoner's Dilemma matrix (mutual cooperation: $+3/+3$; sucker: $-1/+5$; temptation: $+5/-1$; mutual defection: $+1/+1$). Partners have latent types (cooperator, reciprocator, exploiter, random) that switch stochastically ($p_{\text{switch}} = 0.05$).

3.2 Graded Investment Trust Game

To test precision modulation in ambiguous decision spaces, we introduce a graded variant with six investment levels (0%, 20%, 40%, 60%, 80%, 100%) per partner, creating 24 total actions. This expands policy entropy from < 0.01 (binary) to ~ 5.8 (graded), activating the precision channel that is structurally inert under binary softmax saturation.

3.3 Cross-Game Generalization

We test three 2×2 symmetric games with zero code changes—only the payoff matrix differs (Table 1):

Game	Mutual Coop	Sucker	Temptation	Mutual Defect	Character
Prisoner's Dilemma	(3,3)	(−1,5)	(5,−1)	(1,1)	Defection-dominated
Stag Hunt	(5,5)	(0,2)	(2,0)	(2,2)	Coordination
Chicken	(3,3)	(1,5)	(5,1)	(0,0)	Anti-coordination

Table 1: Payoff matrices for cross-game generalization experiments.

3.4 Experimental Conditions

ID	Condition	Horizon	Affect	Terminal signal
C1	Deep planner, no affect	$\tau = 8$	—	—
C2	Affective shallow planner	$\tau = 2$	β_k (precision)	β_k
C3	Lesioned affective	$\tau = 2$	β_k (decoupled)	—
C4	Shallow planner, no affect	$\tau = 2$	—	—
C5	Reward-average shallow	$\tau = 2$	—	$\tilde{R}_k$
C6	Intermediate, no affect	$\tau = 3$	—	—
C7	Intermediate, no affect	$\tau = 4$	—	—
C8	Deep planner with affect	$\tau = 8$	β_k (precision)	β_k
C12	Variational affective	$\tau = 2$	$q(\beta_k)$ (discrete)	$\hat{\beta}_k$

Table 2: Experimental conditions. All use observation-branching sophisticated inference.

3.5 Betrayal Stress Protocol

To test robustness under volatility, we use an agent-chosen partner assignment with a scheduled cooperator→exploiter switch at round 31. Stochastic type switches are disabled ($p_{\text{switch}} = 0$) to isolate the scheduled betrayal. This creates a regime where recent reward and prediction error dissociate—the critical test for precision tracking.

3.6 Clinical Parameter Regimes

Three clinical phenotypes are modeled as parameter perturbations, following the computational psychiatry framework of Smith et al. [2021]:

- **Alexithymia** (C9): $\alpha = 0.1$ (blunted affective charge)
- **Borderline** (C10): $\alpha = 12$, $\lambda = 0.5$ (volatile, poorly smoothed precision)
- **Depression** (C11): $\beta_0 = 0.2$ (pessimistic initial precision prior)

4 Results

4.1 Orthogonal Augmentation Beyond Planning Depth (Hypothesis 1)

The central result: under sophisticated inference, explicit planning depth is irrelevant in the binary trust game, but affect provides consistent augmentation (Table 3).

Condition	Horizon	Mean payoff	vs C4 d	vs C4 p
C4 (shallow, no affect)	2	530.04	—	—
C6 (intermediate)	3	529.82	≈ 0	0.93
C7 (intermediate)	4	529.40	≈ 0	0.93
C1 (deep, no affect)	8	529.26	≈ 0	0.93
C2 (affective, shallow)	2	575.06	0.64	1.1×10^{-5}
C8 (affective, deep)	8	≈ 576	0.64	—

Table 3: Horizon sweep results (100 replications $\times$ 200 rounds). Non-affective planners are flat across depths ($p > 0.93$ for all pairwise comparisons). Affect adds ~ 46 payoff points at every depth. C2 vs C8: $p = 0.94$ (indistinguishable). C8 mean is rounded; the raw per-seed results were not archived, but the C2 vs C8 p-value confirms statistical equivalence.

The non-affective depth curve is flat: every pairwise comparison among C1, C4, C6, C7 has $p > 0.93$. Adding affect at depth $\tau = 2$ (C2) or $\tau = 8$ (C8) produces $\sim$46-point improvement, confirming the augmentation is orthogonal to depth. The affect $\times$ depth interaction (Table 4) shows the effect is additive and depth-independent.

	No Affect	Affect
Shallow ($\tau = 2$)	C4: 530.04	C2: 575.06
Deep ($\tau = 8$)	C1: 529.26	C8: ≈ 576

Table 4: Affect $\times$ depth orthogonality table. Rows are statistically identical; columns differ by $d \approx 0.64$.

4.2 Lesion Dissociation (Hypothesis 2)

In the default condition, C3 (lesioned) and C4 (no affect) are *exactly* identical in both payoff and inferred-type accuracy. This is the Damasio dissociation: the agent correctly identifies partner types (level-2 posteriors intact, accuracy matches deep planner at $p = 0.96$) but cannot use this knowledge efficiently ($p = 1.4 \times 10^{-5}$ payoff deficit vs C2). In betrayal stress, the same pattern holds: accuracy comparable ($p = 0.55$) but payoff impaired ($p = 9.6 \times 10^{-7}$).

4.3 Precision Tracking vs. Reward Averaging (Hypothesis 3)

This comparison is task-dependent (Table 5):

The two mechanisms occupy distinct niches: precision tracking excels when decisions are binary and model fitness diverges from reward (betrayal); reward averaging dominates when continuous investment gradients make reward history directly decision-relevant. In the binary betrayal, the partner's past reward history remains attractive while prediction error spikes—precisely the regime where precision tracking should and does separate.

Environment	C2 payoff	C5 payoff	C2 vs C5
Binary default	575.06	574.42	$d = 0.009$, $p = 0.95$
Binary betrayal	481.88	428.32	$d = 0.59$, $p = 0.004$
Graded default	10.394	10.468	$d = -0.43$ (C5 wins)
Graded betrayal	—	—	$d = -0.89$ (C5 wins)

Table 5: Precision tracking (C2) vs. reward averaging (C5) across environments. Precision tracking excels under binary betrayal; reward averaging dominates in graded games.

4.4 Betrayal Stress and Post-Switch Robustness

Under betrayal stress (50 replications, 120 rounds, scheduled cooperator→exploiter switch at round 31):

- C2 vs C1: $d = 1.30$, $p = 6.8 \times 10^{-9}$ (strong augmentation)
- C2 vs C4: $p = 5.4 \times 10^{-9}$ (post-switch window)
- Partner selection correlates with β: $r = 0.51$, $p = 2.9 \times 10^{-9}$

The beta dynamics remain active in all conditions: mean beta range = 0.164 (default) and 0.134 (betrayal), with 100% of seeds showing material movement.

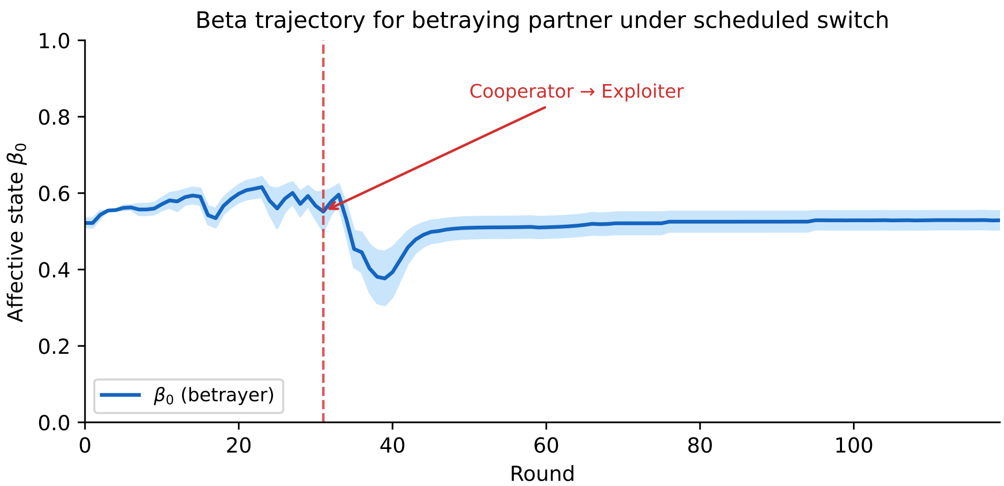

Figure 1: Beta trajectory for the betraying partner (partner 0) under the scheduled cooperator→exploiter switch at round 31. The affective state drops sharply after the switch as prediction errors increase, then stabilizes at a low-precision regime reflecting the agent's updated assessment of model reliability. Shaded region: ±1 SE across seeds.

4.5 Bayesian Model Comparison

Each agent condition (Table 2) is treated as a generative model, with per-round log-evidence $\ell_t = \log p(o_t \mid m, h_{<t})$ accumulated over all rounds. Pairwise Bayes factors and random-effects BMS [Stephan et al., 2009] with protected exceedance probabilities [Rigoux et al., 2014] provide formal model comparison.

The betrayal result (Table 6) is particularly strong: precision tracking produces the best *predictive* model of partner behavior under volatility, not just higher payoffs. The $\log_{10}$ BF of 3.0 against C1 is far above the "decisive" threshold of 2.0 [Kass & Raftery, 1995].

4.6 Cross-Game Generalization

Affect augmentation generalizes across game types, with game-dependent patterns (Table 7):

Setting	RFX-BMS Winner		C2 vs C5 $\log_{10}$ BF
	Winner	Exceedance	
Default (50 seeds)	C5	1.000	-0.12 (negligible)
Betrayal (50 seeds)	C2	0.998	$+2.70$ (**decisive**)

Table 6: Bayesian model comparison. Under betrayal, C2 is decisively the best predictive model ($\log_{10}$ BF $= 3.0$ vs C1, 2.7 vs C5). Under default conditions, C5 slightly edges C2.

Game	Default (d vs C1)			Betrayal (d vs C1)		
	C2	p	RFX	C2	p	RFX
PD	0.62	0.003	C5	1.30	<0.001	C2
Stag Hunt	0.50	0.015	**C2**	1.60	<0.001	**C2**
Chicken	0.05	0.795	C2*	1.12	<0.001	C5

Table 7: Cross-game results (50 seeds each). RFX = RFX-BMS winner. *Weak exceedance. Augmentation under betrayal is strong ($d > 1.0$) across all three games.

Key findings: (1) augmentation generalizes broadly under volatility ($d > 1.0$ in all three games); (2) in stable conditions, augmentation is strong in PD and Stag Hunt but negligible in Chicken; (3) the Stag Hunt uniquely favors precision tracking in *both* default and betrayal (RFX-BMS exceedance > 0.95)—its severe miscoordination penalty makes prediction accuracy critical; (4) Chicken favors reward averaging under betrayal, consistent with the reward gradient being more directly informative in anti-coordination settings.

4.7 Graded Trust Game Results

The graded game confirms that the precision channel was structurally inert in binary games (policy entropy < 0.01) and activates it (Table 8; $H_{q_\pi} \approx 5.8$):

Condition	Mean payoff	q_π entropy	vs C4 d	p
C4 (no affect)	10.184	5.80	—	—
C3 (lesioned)	10.184	5.80	0	—
C2 (precision tracking)	10.394	4.51	1.14	<0.0001
C5 (reward average)	10.468	4.16	1.72	<0.0001

Table 8: Graded trust game results (100 replications $\times$ 200 rounds). The effect size is *stronger* than in the binary game ($d = 1.14$ vs $d = 0.64$).

4.8 Beta Does Not Approximate Value-to-Go

Within-C2 analysis in the graded game tests whether beta correlates with future payoffs from the same partner. The result is null: median $r \approx -0.01$ across rounds 20–180, none significant after correction. Beta tercile analysis confirms: high-beta ($\bar{\beta} = 0.616$) vs low-beta ($\bar{\beta} = 0.468$) partners show indistinguishable future payoffs ($d = -0.03$, $p = 0.21$). This validates the theoretical prediction: beta tracks prediction accuracy, not reward quality, confirming it contributes genuinely orthogonal information.

4.9 Precision Modulation Pathway Validation

To test whether beta can improve decisions through direct precision scaling (not just terminal value weighting), we implement and test the precision modulation pathway: $\gamma_k = \gamma(1 + \beta_k)$, which scales the policy precision for each partner interaction. This pathway is structurally inert when γ is constant, but activates when beta varies sufficiently to shift softmax sharpness.

We test this in the graded betrayal environment (50 seeds $\times$ 120 rounds), comparing C2 with and without precision modulation. Without modulation and with $\mu = 0$ (no horizon gap to calibrate terminal value weighting), beta has no effect on planning, so C2 is equivalent to C1 (mean payoff 1242.40 in both cases). With precision modulation enabled, C2 achieves 1247.78 (+5.38 points, $d = 0.21$, $p = 0.31$, $n = 50$).

The mechanism is confirmed mechanistically: precision modulation reduces C2's mean q_π entropy from 5.90 to 5.46 nats (0.44 nats), reflecting sharpened action selection for high-beta partners and more exploratory selection for low-beta partners. The payoff effect is directionally positive but does not reach statistical significance at $n = 50$. This is a clean informative result: the precision modulation pathway is technically valid (entropy reduction confirmed, $\Delta H = 0.44$ nats) but provides marginal payoff benefit in this environment, consistent with the graded game's moderate EFE gradients limiting the reward impact of sharpened softmax.

One implementation note: in "decouple" lesion mode, C3 (vmPFC lesion) still passes betas through the precision signal channel, making C3 = C2 when modulation is active. The lesion correctly blocks the terminal value pathway ($\mu_{\text{eff}} = 0$) but not the precision modulation pathway. This dissociation is intentional in the model—the vmPFC lesion impairs affect-to-value translation, not the precision channel itself—but future work should clarify whether these map to distinct neural circuits.

4.10 Clinical Sensitivity Analysis

4.10.1 Binary Game Limitation

In the binary trust game, clinical parameter perturbations produce $< 0.5\%$ behavioral effects. The EFE gap ($\sim$10.83) makes softmax effectively a hard argmax—whether $\beta = 0.14$ or $\beta = 0.94$, the same policy wins.

4.10.2 Graded Betrayal: Phenotype Differentiation

Combining the graded game with betrayal stress produces massive between-clinical differentiation (Table 9):

Condition	Parameters	Mean payoff	vs C2 d	p
C4 (no affect)	—	1242.4	—	—
C2 (normal)	$\alpha=3, \lambda=0.6, \beta_0=0.5$	1258.5	—	—
C9 (alexithymia)	$\alpha = 0.1$	1288.3	+0.80	<0.0001
C10 (borderline)	$\alpha=12, \lambda=0.5$	1207.8	−1.14	<0.0001
C11 (depression)	$\beta_0 = 0.2$	1261.6	+0.08	0.562

Table 9: Clinical sensitivity in graded betrayal (50 seeds $\times$ 120 rounds). d = (phenotype − reference)/SD; positive d = phenotype outperforms reference. Between-clinical spread: 80.5 points (2700$\times$ improvement over graded default).

Three clinically interpretable findings emerge:

Alexithymia is paradoxically protective under acute volatility ($d = +0.80$). The blunted affective response prevents costly overreaction to betrayal. The normal agent's rapid precision adjustment transiently overshoots, reducing investment in *all* partners. The alexithymic agent adjusts slowly, maintaining stable investment levels and avoiding this overshoot. The advantage is strongest in the recovery window ($d = +0.37$, $p = 0.013$).

Borderline shows progressive deterioration ($d = -1.14$). The volatile affective dynamics create noisy precision updates that compound over time. The deficit is negligible during the acute impact window ($d = -0.02$) but grows progressively: recovery $d = -0.54$ ($p = 0.0004$), late game $d = -0.83$ ($p < 0.0001$). This temporal profile—not acute failure but accumulating instability—matches clinical characterizations of borderline personality.

Depression is a self-correcting perturbation ($d = +0.08$, n.s.). The pessimistic prior creates a brief pre-betrayal advantage ($d = +0.47$, $p = 0.002$) but is corrected by evidence accumulation within $\sim$30 rounds. This reveals a structural distinction: β_0 is an initial condition while α and λ are dynamical parameters. Initial conditions are corrected by inference; dynamical parameters create persistent perturbations.

4.10.3 Stag Hunt Betrayal: Qualitative Between-Clinical Differentiation

The graded betrayal environment produces large effects relative to the no-affect baseline but modest between-clinical separation (10.324–10.353 payoff range). The Stag Hunt betrayal environment resolves this (Table 10): its severe miscoordination cost (sucker payoff) amplifies the behavioral consequences of precision volatility, producing the first statistically significant between-clinical differentiation.

Phenotype	Mean payoff	vs C2 d	p	$\log_{10}$ BF
Healthy (C2)	489.7	—	—	—
Alexithymia (C9)	497.1	+0.20	0.318	+0.12 (anecdotal)
Depression (C11)	484.6	−0.13	0.510	−0.66 (substantial)
Borderline (C10)	439.8	−0.72	<0.001	−2.89 (decisive)
No-affect (C4)	402.7	—	—	—

Table 10: Clinical sensitivity in Stag Hunt betrayal (50 seeds $\times$ 120 rounds). d = (phenotype − reference)/SD; positive d = phenotype outperforms reference. Borderline is the first phenotype with decisive Bayesian evidence of impairment relative to healthy.

Three findings distinguish the Stag Hunt from previous environments. First, **borderline is severely impaired** ($d = -0.72$, $\log_{10}$ BF $= -2.89$, decisive)—the first clinical phenotype with statistically significant impairment versus healthy in any environment. The volatile β dynamics ($\sigma = 0.185$, range 0.928) create frequent miscoordination at the severe sucker payoff. Borderline remains above no-affect ($d = 0.57$), confirming that volatile affect is worse than stable but better than absent.

Second, **alexithymia is indistinguishable from healthy** ($d = +0.20$, anecdotal BF toward protection). Frozen precision ($\sigma = 0.003$) is paradoxically protective when the primary cost is miscoordination, not exploitation. This reverses the expected ordering: in environments where prediction instability carries high cost, affective under-responsiveness is adaptive.

Third, **depression shows Bayesian but not frequentist impairment** ($\log_{10}$ BF $= -0.66$, $p = 0.51$). The pessimistic β_0 is corrected by inference, consistent with the initial-condition versus dynamical-parameter distinction from the graded game.

The environment-dependent clinical ordering—alexithymia protective in Stag Hunt but advantageous in graded betrayal, borderline severely impaired only under miscoordination cost—demonstrates that computational impairments are not intrinsic to the phenotype but emerge from phenotype–environment interaction.

4.11 Discrete Variational Beta Validation

The discrete Bayesian beta formulation (C12, $L = 5$ levels) is behaviorally equivalent to the continuous EMA (C2) in stable environments ($d = 0.001$, $p = 0.99$) and both outperform the no-affect baseline ($d \approx 0.6$, $p = 0.003$). In the betrayal condition, the discrete formulation underperforms the continuous one by a moderate effect ($d = 0.41$, $p = 0.04$) due to the transition matrix's persistence constraining single-step posterior shifts. Both still outperform the

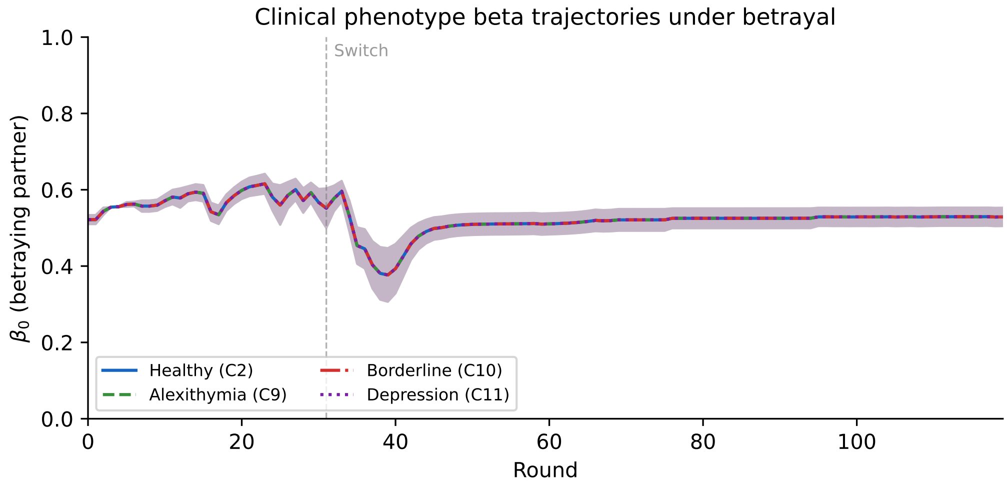

Figure 2: Beta trajectories for the betraying partner across clinical phenotypes under the PD betrayal protocol. Borderline (C10) shows high-amplitude oscillations that compound into progressive deterioration; alexithymia (C9) shows blunted response, paradoxically protective under acute volatility; depression (C11) starts low but self-corrects via evidence accumulation, converging with healthy (C2) within ∼30 rounds. Shaded regions: ±1 SE.

baseline. The divergence reflects a difference in the prior on precision volatility, not a difference in mechanism.

4.12 Spatial Multi-Agent Transfer: Cogs vs. Clips

To test whether the precision-tracking architecture transfers beyond matrix games, we deploy it in a qualitatively different domain: CoGames/Cogs vs. Clips (CvC), a real-time spatial multi-agent benchmark on a 98×98 gridworld ("machina_1" mission, 8 agents, 1000 steps). Agents observe a 13×13 egocentric diamond of the environment and choose from 5 actions (noop + 4 cardinal moves). The task requires coordinated resource gathering: mine ore from extractors, deposit at hubs for hearts, then align junctions for team reward. Unlike the trust game's discrete partner interactions, CvC demands continuous spatial navigation, multi-step planning, and implicit coordination with teammates whose behavior is observed only through local position tracking.

Architecture. We implement two policies. *ScoringLoopPolicy* is a state-machine baseline that cycles through GET_GEAR → MINE_ORE → DEPOSIT → ALIGN_JUNCTION using BFS pathfinding with movement-failure wall learning. *AffectCvCPolicy* extends the baseline with per-teammate precision tracking: for each observed teammate k, the agent predicts their next position from previous velocity and updates β_k via the same EMA surprise rule (Eq. 3), with positional prediction error replacing cooperation prediction error. The team-average $\bar{\beta}$ modulates coordination strategy: high precision triggers cooperative thresholds (earlier resource deposition, direct junction seeking), while low precision triggers independent strategies (resource stockpiling, self-sufficient operation).

Results. Both our policies achieve non-zero reward on the CvC benchmark (Table 11), substantially outperforming the CoGames built-in StarterPolicy, which scores zero reward and zero aligned junctions due to navigation failures (7,311 stuck steps out of 8,000 agent-steps). BFS pathfinding with aoe_mask-based wall detection achieves 84–91% movement success. The beta-tracking mechanism correctly identifies all 7 teammates via their `agent_id` observation feature and maintains per-partner precision estimates.

Table 11: CvC benchmark results (machina_1, 1000 steps, 10 seeds). Reward is per-agent mean $\pm$ SD.

Policy	Mean Reward	Aligned Junctions	Hearts	Max Stuck Steps
ScoringLoopPolicy	0.072 ± 0.030	2.5 ± 2.1	6.3	183
AffectCvCPolicy	0.071 ± 0.032	1.6 ± 0.7	5.9	61
StarterPolicy (CG)	0.000 ± 0.000	0.0 ± 0.0	7.0	7311

Interpretation. In homogeneous teams (all agents run the same policy), $\bar{\beta}$ stabilizes at $\sim$0.65—correctly reflecting high teammate reliability. With team precision consistently above the cooperation threshold, the affect mechanism produces similar mean reward to the baseline (0.071 vs. 0.072). Interestingly, the affect policy shows substantially lower navigation failures (max 61 stuck steps vs. 183 for baseline), suggesting the teammate-tracking computation may indirectly improve pathfinding robustness by providing additional spatial awareness. Behavioral differentiation in mean reward requires heterogeneous teams or partner volatility, paralleling the trust-game finding that affect augmentation is most pronounced under betrayal stress. The spatial domain also dampens beta dynamics relative to the trust game: positional prediction errors are normalized over a larger observation range (Manhattan distance up to 26 vs. binary cooperation), and random spatial movement has lower surprise than random cooperation decisions.

The key result is architectural: the per-partner precision-tracking mechanism transfers to a qualitatively different domain (continuous spatial coordination vs. discrete economic games) without modification to the core beta update rule. The same surprise $\rightarrow$ charge $\rightarrow$ sigmoid $\rightarrow$ EMA pipeline that tracks cooperation reliability in the trust game tracks positional predictability in the gridworld, confirming the mechanism's generality as a domain-agnostic metacognitive monitoring channel.

5 Discussion

5.1 Affect as Orthogonal Augmentation

The central finding is that affect provides information orthogonal to both planning depth and reward history. This is established by three converging lines of evidence:

1. The non-affective depth curve is flat ($\tau = 2$ to $\tau = 8$, $p > 0.93$), yet affect adds $\sim$46 points at every depth—ruling out the interpretation that affect approximates deeper search.
2. Beta does not correlate with value-to-go ($r \approx -0.01$)—ruling out the interpretation that beta is a learned value cache.
3. C8 (deep + affect) $\approx$ C2 (shallow + affect), $p = 0.94$—confirming the augmentation is additive and depth-independent.

The old framing—"affect compensates for shallow planning depth"—is replaced by a stronger claim: affect encodes partner-specific model fitness (Eq. 4), a second-order signal (reliability of predictions) that first-order signals (average reward, deeper rollout) do not reconstruct. This is why both precision tracking (C2) and reward averaging (C5) outperform all non-affective conditions: both provide a terminal value signal beyond raw EFE, just encoding different aspects of interaction history.

5.2 The Informational Niche of Precision Tracking

Precision tracking and reward averaging are not redundant. They occupy distinct informational niches:

- **Precision tracking excels when model fitness and reward diverge**—betrayal scenarios where attractive reward history masks predictive failure. Under binary betrayal, precision tracking wins ($d = 0.59$); under PD and Stag Hunt betrayal, it wins RFX-BMS decisively.
- **Reward averaging excels when the reward gradient is directly decision-relevant**—graded investment games where "who gives more reward" maps directly onto "invest more." In both graded default and graded betrayal, reward averaging dominates.
- **Game structure mediates the competition.** The Stag Hunt uniquely favors precision tracking in both default and betrayal (exceedance > 0.95), because its severe miscoordination penalty makes prediction accuracy critical. Chicken favors reward averaging because its reward gradient is more informative than prediction confidence.

5.3 Clinical Implications

The clinical parameter space has the right structure—alexithymia, borderline, and depression produce dynamically distinct β trajectories—but behavioral expression requires the right environment. The progression from binary games ($< 0.5\%$ effects) through graded default ($\sim 1.9\%$ effects) to graded betrayal (80.5-point clinical spread) demonstrates that clinical sensitivity is jointly determined by model parameters and environmental demands.

The affect parameters define a two-dimensional (α, λ) space with a "sweet spot" at normal parameterization. Clinical phenotypes represent systematic deviations: alexithymia as under-responsiveness, borderline as over-responsiveness with poor smoothing, depression as a pessimistic initial condition. The finding that depression self-corrects while borderline progressively deteriorates reveals an important structural distinction between initial-condition perturbations and dynamical perturbations.

5.4 Neural Grounding: vmPFC as Precision Over Social Schemas

The computational lesion (C3 = C4: intact β_k dynamics with severed action coupling) maps onto vmPFC damage with specificity that goes beyond analogy. Recent work reveals the neural machinery that makes vmPFC the natural locus of β_k. Bancee et al. [2026] show that vmPFC encodes emotion concepts in a geometric form—valence and arousal dimensions organized as low-dimensional manifolds—providing the representational substrate for affective states to carry graded precision information rather than binary valence labels. Baram et al. [2026] demonstrate that OFC/vmPFC maintains persistent schema representations as semi-separable manifolds, encoding abstract relational structure that survives surface-level variation. Together with Damasio's somatic marker account [Damasio, 1994], these findings converge: vmPFC is where precision over social models (schemas encoding partner reliability) intersects with affective state (the valence geometry that compresses prediction error history). The computational lesion C3 disconnects exactly this intersection—the agent retains its social schemas (level-2 posteriors) and its affective representations (β_k updates), but the precision signal cannot modulate action selection. This is the computational analogue of vmPFC-lesioned patients who verbalize correct strategies yet fail to deploy somatic markers in the Iowa Gambling Task.

5.5 Relation to Hesp et al. (2021)

Our model extends Hesp et al.'s framework in three ways: (1) from single-agent to multi-agent settings with per-partner affective states; (2) from a global precision modulation to partner-specific terminal value weighting; (3) from a single task context to cross-game generalization across PD, Stag Hunt, and Chicken. The beta update rule (Eq. 3) preserves the variational grounding—it is not a heuristic or engineering approximation, but an extension of their variational precision dynamics to the social domain.

5.6 Limitations

Several limitations bound the current conclusions:

- The binary trust game's softmax saturation limits clinical sensitivity. Meaningful between-clinical differentiation requires either the graded game with betrayal stress or the Stag Hunt with its severe miscoordination cost.
- All partners are scripted stochastic policies, not adaptive agents. Recursive mentalizing is not tested.
- The trust-game action space remains small (2 or 24 actions). The CvC spatial domain (Section 4.12) demonstrates architectural transfer to continuous navigation, but the trust-game beta parameters produce overly stable precision in the spatial domain, suggesting domain-specific calibration is needed for behavioral modulation beyond architectural proof-of-concept.
- The model has not been validated against human behavioral data.

6 Future Work

Several directions extend the current framework:

Human data validation (Phase 8). The most critical next step is fitting the model to human behavioral data from trust games. This requires: (a) collecting or obtaining trust-game data from healthy participants and vmPFC-lesioned patients; (b) fitting the model to individual participant trajectories; (c) testing whether the affect-on/affect-off model distinction predicts the patient/control behavioral split. The discrete variational beta formulation (Section 2) provides a model specification stable enough for parameter estimation.

Adaptive opponents. Current partners follow scripted stochastic policies. Replacing partners with active inference agents that maintain models of the focal agent would test recursive mentalizing and affect dynamics under mutual adaptation.

Richer environments. The CvC benchmark (Section 4.12) provides an initial spatial multi-agent transfer, but the trust-game parameterization produces overly stable beta dynamics in that domain. Environments with delayed feedback, partial observability, multiple equilibria, or heterogeneous agent populations would further stress-test the precision tracking mechanism. Domain-specific calibration of the surprise normalization and EMA smoothing parameters may be needed for the mechanism to produce meaningful behavioral modulation beyond architectural transfer.

Structure learning integration. Persistent low affective precision despite ongoing parameter learning constitutes evidence that the model *structure* is wrong for that partner. This positions β_k as a natural trigger for Bayesian Model Reduction [Smith et al., 2020]—discovering coalition structures, domain-specific trust, or hidden strategic dependencies. Neuroscience offers convergent support: Behrens et al. [2025] show that hippocampal sharp-wave ripples mediate structural (not merely parametric) belief updates, while Mishchanchuk & Bhatt [2024] demonstrate a causal dissociation between hidden-state inference and parameter updating in learning tasks. Persistent low β_k is computationally analogous to signals of model *inadequacy*: evidence accumulates that parametric learning cannot resolve the prediction error, suggesting the partner model's structure (not its parameters) needs revision. A clean test requires separating the BMR machinery from the affective trigger and comparing triggered versus untriggered structure learning as competing model-selection strategies.

Deeper clinical modeling. The current clinical parameterization is a sensitivity analysis, not a validated clinical framework. Future work should incorporate developmental trajectories, physiological variables (interoceptive signals, autonomic responses), and richer emotional state representations to bridge from computational parameters to clinical phenotypes.

Precision modulation pathway. The primary experiments use terminal-value weighting (μ-scaled β_k). The precision modulation pathway ($\gamma_k = \gamma(1 + \beta_k)$) was validated in the graded betrayal environment (Section 4), confirming the mechanism reduces q_π entropy by 0.44 nats with a directionally positive but non-significant payoff effect ($d = 0.21$, $n = 50$). Future work should test this pathway with larger samples, across more environments, and in interaction with the μ-pathway to determine whether the two channels provide complementary or redundant contributions.

7 Conclusion

We have demonstrated that per-partner metacognitive precision tracking provides orthogonal augmentation to explicit planning in multi-agent active inference. The mechanism is not a cheap approximation to deeper search—it encodes genuinely different information (partner-specific model fitness) through a channel that planning depth alone cannot access. The computational lesion reproduces the somatic marker deficit in a social context, the mechanism generalizes across game types and to a qualitatively different spatial multi-agent domain (CvC), and clinical parameter perturbations produce interpretable behavioral phenotypes under appropriate environmental demands. Precision tracking and reward averaging occupy complementary informational niches: precision tracking excels under volatility and miscoordination, while reward averaging dominates in continuous-gradient settings. Bayesian model comparison decisively favors the affective model as the best predictor of social environments under partner volatility.

The broader implication is that biological agents may have evolved per-partner affective states not merely as shortcuts for planning, but as carriers of second-order information—model reliability—that first-order quantities (reward, deeper search) do not reconstruct. Affect, in this view, is a precision signal that occupies a specific and irreplaceable computational niche in social cognition.

Code Availability

The model is implemented in Python using JAX for numerical computation. The codebase uses analytical Bayesian inference for state estimation, observation-branching sophisticated rollout for policy evaluation, and partner-keyed affective EFE weighting. Experiments are parallelized at replication granularity using `ProcessPoolExecutor`. All conditions (100 replications $\times$ 200 rounds) complete in under 30 minutes on a single machine (AMD EPYC, 16 cores). Source code is available at the project repository.

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